



Cluster of differentiation 14 and toll-like receptor 4 are involved in the anti-inflammatory effects of tyrosol

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ABSTRACT

We investigated effects of tyrosol on upstream pathway regulating lipopolysaccharide (LPS)-induced inflammatory response, including cluster of differentiation 14 (CD14)-mediated LPS-macrophage binding, toll-like receptor 4 (TLR4)/myeloid differentiation factor 2 (MD2) expression and subsequent myeloid differentiation primary response gene 88 (MyD88) and TIR-domain-containing adapter-inducing interferon- β (TRIF) recruitments. Flow cytometry revealed that, compared with LPS-treated RAW264.7 cells (LPS group), those treated with LPS and tyrosol (1.2 mM) [LPS + T(1.2) group] demonstrated lower LPS-macrophage binding and membrane-bound CD14 (mCD14) and TLR4/MD2 expression (all $p < 0.05$). Immunoprecipitation/immunoblotting assay revealed lower MyD88 and TRIF concentrations in the LPS + T(1.2) group than in the LPS group (both $p < 0.05$). Soluble CD14 concentration was higher in the LPS + T(1.2) group than in the LPS group ($p = 0.013$); protease inhibition counteracted this effect. Thus, tyrosol inhibits LPS-macrophage binding, mCD14 and TLR4/MD2 expression, and MyD88 and TRIF recruitment, all possibly by enhancing protease-mediated mCD14 cleavage.

1. Introduction

Lipopolysaccharide (LPS), also called the gram-negative endotoxin, is a major component of the outer membrane of gram-negative bacteria that can activate major inflammatory signalling pathways in macrophages (Manthey & Vogel, 1994). The pathways begin with LPS-macrophage binding mediated by the accessory proteins, namely the circulating LPS-binding protein (LBP) and the cell surface pattern recognition receptor cluster of differentiation 14 (CD14) (Park & Lee, 2013; Schumann, 1992). LBP is generally believed to present LPS to

CD14, which then facilitates the chaperone of monomeric forms of LPS and enhances LPS transfer to the toll-like receptor 4 (TLR4)/myeloid differentiation factor 2 (MD2) complex (Fitzgerald, Rowe, & Golenbock, 2004; Jerala, 2007; Pålsson-McDermott and O'Neill, 2004; Park & Lee, 2013; Schumann, 1992). CD14 exists in two forms: membrane-bound CD14 (mCD14), which anchors to the membrane by a glycosylphosphatidylinositol tail, and soluble CD14 (sCD14) (Kirkland & Viriyakosol, 1998), produced through either the proteolytic cleavage of mCD14 (i.e., shedding of CD14) or direct secretion from intracellular vesicles (Pham, 2008; Tada et al., 2002). The effects of mCD14 and

Abbreviations: CD14, cluster of differentiation 14; ELISA, enzyme-linked immunosorbent assay; FITC-LPS, lipopolysaccharide labelled with fluorescein isothiocyanate; IL-6, interleukin-6; LBP, lipopolysaccharide-binding protein; LPS, lipopolysaccharide; MAPKs, mitogen-activated protein kinases; mCD14, membrane-bound CD14; MD2, myeloid differentiation factor 2; MFI, mean fluorescence intensity; MyD88, myeloid differentiation primary response gene 88; NF- κ B, transcription factors nuclear factor- κ B; PBS, phosphate-buffered saline; PI, protease inhibitors; sCD14, soluble CD14; TRIF, TIR-domain-containing adapter-inducing interferon- β ; TLR4, toll-like receptor 4

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sCD14 are complex. mCD14 is responsible for exposure of LPS to TLR4/MD2 (Berenger et al., 2011). However, sCD14 may suppress responses to LPS by removing LPS from mCD14 (Berenger et al., 2011). The crucial role of CD14 has been highlighted previously: CD14 is an essential factor enabling the LPS receptor to switch into the “full activation” mode (Jiang et al., 2005).

The TLR4/MD2 complex comprises an extracellular adaptor protein MD2 and a transmembrane receptor TLR4 (Park & Lee, 2013; Schumann, 1992). The binding of LPS to TLR4/MD2 induces TLR4/MD2 activation, including TLR4/MD2 dimerization and subsequent TLR4 endocytosis (Fitzgerald et al., 2004; Jerala, 2007; Pålsson-McDermott and O'Neill, 2004; Park & Lee, 2013; Schumann, 1992). Dimerization of TLR4/MD2 at the cell surface induces recruitment of myeloid differentiation primary response gene 88 (MyD88) (Kagan & Medzhitov, 2006). TLR4 endocytosis into endosomes then induces recruitment of TIR-domain-containing adapter-inducing interferon- β (TRIF) (O'Neill et al., 2003; Yamamoto et al., 2003). The MyD88-mediated early response then activates transcription factors nuclear factor- κ B (NF- κ B) and mitogen-activated protein kinases (MAPKs) and subsequently induces upregulation of inflammatory molecules (e.g., cytokines) (Christman, Lancaster, & Blackwell, 1998; Kristof, Marks-Konczalik, & Moss, 2001). The subsequent TRIF-mediated response activates interferon regulatory factor 3 and then induces interferon- β production followed by NF- κ B activation (Kagan et al., 2008).

Tyrosol [2-(4-hydroxyphenyl)ethanol] is a polyphenol presents in natural sources, including olive oil, red wine, and white wine (Covas et al., 2003; Miró-Casas et al., 2001). Tyrosol has potent antioxidant and anti-inflammatory properties (Sato et al., 2016; Serra, Deiana, Spencer, & Corona, 2017). It can inhibit LPS-induced cytokine production in human peripheral blood mononuclear cells (Giovannini et al., 2002) and mitigate inflammatory response, inhibit organ injuries, and increase survival in rodents with endotoxemia (Kim et al., 2017; Lu et al., 2013). Moreover, the involvement of NF- κ B and MAPKs mediate the anti-inflammation effects of tyrosol (Sato et al., 2016; Wang et al., 2017). These data highlight the effectiveness of tyrosol in modulating LPS signal transduction pathways. CD14, as mentioned, plays a crucial role in mediating LPS-induced TLR4/MD2 activation. However, whether tyrosol modulates the aforementioned upstream pathways remains unclear. To further elucidate this, we conducted a cellular study by hypothesising that tyrosol inhibits LPS–macrophage binding, mitigates mCD14 activation, increases sCD14 expression, inhibits TLR4/MD2 expression, and reduces MyD88 and TRIF recruitment in LPS-treated macrophages. With previous data supporting the beneficial effects of tyrosol, the current data can provide detailed information regarding the upstream regulatory pathways involved in mediating the anti-inflammatory effects of tyrosol. Findings from this study will help advance knowledge in this area.

In summary, this study investigated the effects of tyrosol on modulating the upstream pathways regulating LPS-induced inflammatory response involving CD14, TLR4/MD2, MyD88, and TRIF.

2. Materials and methods

2.1. Cell culture and activation with LPS, tyrosol administration, and cell viability

RAW264.7 cell line, a widely used immortalised murine macrophage-like cell line, was grown and incubated as per our previously published protocol (Huang, Tsai, Kai, Yang, & Huang, 2006). Only the cells that had undergone < 20 passages were used in this experiment.

The cells were then treated with 100 ng/mL LPS (*Escherichia coli* Serotype 0127:B8 endotoxin; Sigma-Aldrich, St. Louis, MO, USA) to induce expression of the molecules under investigation. LPS dose was determined according to our previous report (Chen, Tsai, & Huang, 2012). For LPS–macrophage binding assay through flow cytometry, another set of RAW264.7 cells was treated with LPS labelled with

fluorescein isothiocyanate (FITC-LPS; 1000 ng/mL; Sigma-Aldrich). FITC-LPS dose was selected according to our preliminary data, such that the fluorescence signal was detectable. Our preliminary data also revealed that the interleukin 6 (IL-6) concentration in RAW264.7 cells treated with 1000 ng/mL LPS-FITC was approximately 80% of that in cells treated with 1000 ng/mL LPS and approximately threefold higher than that in cells treated with 100 ng/mL LPS. Tyrosol (0.3, 0.6, or 1.2 mM; 98% purity, 188825, Sigma-Aldrich) was administered to RAW264.7 cells 60 min before LPS. The administration time and doses of tyrosol were determined according to previous reports (Kim et al., 2017; Wang et al., 2017). Tyrosol was present in the LPS incubation time in RAW264.7 cells treated with both LPS and tyrosol.

The cell viability of each group was assessed through fluorometric detection of living cell numbers using a commercial kit (Cell Counting Kit-8, CCK-8; Sigma-Aldrich). In brief, cells were seeded in 96-well plates (2×10^4 cells per well; Corning Life Sciences, Oneonta, NY, USA) and incubated for 24 h at 37 °C with 5% CO₂. Then, RAW264.7 cells were treated with 100 ng/mL LPS alone (LPS group) or in combination with 0.3, 0.6, or 1.2 mM tyrosol [Sigma-Aldrich; denoted as LPS + T(0.3), LPS + T(0.6), and LPS + T(1.2) groups, respectively]. LPS and tyrosol were administered as above-mentioned. In addition, another two groups of RAW264.7 cells were treated with phosphate-buffered saline (PBS; Sigma-Aldrich) and tyrosol (1.2 mM) to serve as the vehicle and tyrosol controls, respectively. These two groups were denoted as the PBS and T(1.2) groups, respectively. Tyrosol was dissolved in 5 μ L of dimethyl sulfoxide (DMSO; Sigma-Aldrich); thus, 5 μ L of DMSO was also added in the cells not receiving tyrosol. After 18 h of LPS administration or at the comparable time point in groups without LPS administration, the culture medium of each well was removed. Then, 10 μ L CCK-8 solution (Sigma-Aldrich) in 100 μ L fresh culture medium was added per well and the plates were incubated for 4 h at 37 °C with 5% CO₂. Then, the absorbance was measured at 450 nm using a microplate reader, as per the manufacturer's protocol.

2.2. Measuring IL-6 secretion by using enzyme-linked immunosorbent assay

To determine the effects of tyrosol on IL-6 production, RAW 264.7 cells were seeded in 6-well plates (5×10^5 cells per well; Corning) and incubated at 37 °C with 5% CO₂ for 24 h. The cells were then treated with PBS, 1.2 mM tyrosol, 100 ng/mL LPS alone, 100 ng/mL LPS in combination with 0.3, 0.6, or 1.2 mM tyrosol [denoted as PBS, T(1.2), LPS, LPS + T(0.3), LPS + T(0.6), and LPS + T(1.2) groups, respectively]. LPS and tyrosol were administered as mentioned in Section 2.1. After 18 h of LPS administration or at the comparable time point in groups without LPS administration, the cells were harvested. The freshly harvested culture medium was centrifuged at 500g for 5 min, and the supernatant was collected. The IL-6 concentration of the supernatant from each group was then assayed using enzyme-linked immunosorbent assay (ELISA). A commercial IL-6 ELISA kit (Pierce Biotechnology, Inc., Rockford, IL, USA) was used, and analyses were performed as per the manufacturer's protocol.

2.3. Measuring NF- κ B and MAPKs expression by using immunoblotting assay

RAW264.7 cells were seeded in 6-cm culture dishes (3×10^6 cells per dish; Corning) and incubated at 37 °C with 5% CO₂ for 24 h. Then, RAW264.7 cells were treated with PBS, 1.2 mM tyrosol, 100 ng/mL LPS, or 100 ng/mL LPS and 1.2 mM tyrosol [i.e., PBS, T(1.2), LPS, and LPS + T(1.2) groups, respectively]. LPS and tyrosol administrations were performed as mentioned in Section 2.1. After 30 min of LPS administration or at the comparable time point in groups without LPS administration, the cells were harvested and the whole cell lysates and the nuclear and cytosolic extracts of the cell cultures were prepared according to our previous reports (Chen et al., 2012; Tsai et al., 2006). In brief, cells went through the process of washing, scraping, and

centrifugation (1500g for 5 min). Then, cell pellet was re-suspended in cell lysis buffer [5 mL of 10 mM HEPES (pH 7.9), 1.5 mM MgCl₂, 10 mM KCl, 0.5 mM dithiothreitol, and 0.2 mM phenylmethylsulfonyl fluoride] and centrifuged again (1500g for 5 min). Cells were re-suspended, allowed to swell on ice (10 min) and then homogenized. Homogenates were centrifuged (3300 g, 15 min, 4 °C) and the supernatants and pellets were collected for cytosolic extracts and nuclear extracts, respectively. For preparation of nuclear extracts, the pellets were re-suspended, homogenized in nuclear extraction buffer [3 volumes of 20 mM HEPES (pH 7.9), 1.5 mM MgCl₂, 400 mM KCl, 0.5 mM dithiothreitol, 0.2 mM phenylmethylsulfonyl fluoride, and 25% glycerol], stirring on ice (30 min) and centrifuged (89,000g for 30 min). The supernatant were collected and concentrated by using a concentrator (Microcon 10 concentrator; Millipore Corporation; Burlington, MA, USA) and centrifugation (14,000 g, 3 h, 4 °C). For preparation of cytosolic extracts, the supernatants were mixed with cytoplasmic extraction buffer [30 mM HEPES (pH 7.9) at 4 °C, 140 mM KCl, 3 mM MgCl₂] and centrifuged (89,000 g for 1 h). Then, the supernatants were collected and also concentrated by using a concentrator (Microcon 10 concentrator; Millipore) and centrifugation (14,000 g, 1 h, 4 °C). The protein concentration of each sample (whole cell lysates, nuclear extracts, and cytosolic extracts) was measured using bicinchoninic acid assay (BCA protein assay kit; Pierce Biotechnology, Inc., Rockford, IL, USA). Then, equal amount protein (65 µg) of each sample (nuclear and cytosolic extracts) was loaded to Tris-glycine precast polyacrylamide gels (7.5%, Bio-Rad Laboratories, Hercules, CA, USA) and separated by electrophoresis followed by transfer to nitrocellulose membranes (Bio-Rad).

NF-κB activation is evident by NF-κB nuclear translocation and phosphorylation of inhibitor-κB (I-κB) (Christman et al., 1998). For NF-κB nuclear translocation assay, the membranes of nuclear extracts were incubated with primary antibody of phosphorylated NF-κB p65 (Phospho-NF-κB p65 [Ser536], 1:500 dilution; Cell Signaling Technology, Inc, Danvers, MA, USA) or Histone H3 (internal standard, 1:500 dilution; Cell Signaling). For I-κB phosphorylation assay, the membranes of cytosolic extracts were incubated with primary antibody of phosphorylated I-κBα (Phospho-I-κBα [Ser32], 1:1000 dilution; Cell Signaling) or Actin (internal standard; 1:5000 dilution; Millipore). MAPK activation is evident by phosphorylation of MAPKs (Kristof et al., 2001). For MAPKs phosphorylation assay, the membranes of cytosolic extracts were incubated with primary antibody of phosphorylated extracellular regulated kinase (ERK) (1:500 dilution, polyclonal p-ERK1/2 antibody; Santa Cruz Biotechnology, Inc.; Santa Cruz, CA), phosphorylated c-jun N-terminal kinase (JNK) (1:500 dilution, polyclonal p-JNK1/2 antibody, Santa Cruz), phosphorylated p38 MAPK (1:200 dilution, polyclonal p-p38 MAPK antibody, Santa Cruz) or Actin (1:5000 dilution; Millipore). Then, all membranes were incubated with the secondary antibody solution of horseradish peroxidase conjugated anti-mouse IgG antibody, (Amersham Pharmacia Biotech, Inc., Piscataway, NJ, USA) followed by detection with a chemiluminescence system (ECL plus kit; Amersham). Densitometry technology (ImageJ, freely available at <http://rsb.info.nih.gov/ij/>) was used to quantify densities of the protein bands.

2.4. Assays of LPS-macrophage binding and surface expression of mCD14 and TLR4/MD2 through flow cytometry

RAW264.7 cells were seeded in 6-well plates (1 × 10⁶ cells per well; Corning) and incubated at 37 °C with 5% CO₂ for 24 h. Then, RAW264.7 cells were treated with PBS, 1.2 mM tyrosol, 1000 ng/mL FITC-LPS, or 1000 ng/mL FITC-LPS and 1.2 mM tyrosol [i.e., PBS, T(1.2), LPS, and LPS + T(1.2) groups, respectively] to facilitate assaying LPS-macrophage binding. Also after 24 h of incubation, RAW264.7 cells were then treated with PBS, 1.2 mM tyrosol, 100 ng/mL LPS, or 100 ng/mL LPS and 1.2 mM tyrosol [denoted as the PBS, T(1.2), LPS, and LPS + T(1.2) groups, respectively] to facilitate assaying surface expression of mCD14 and TLR4/MD2. LPS and tyrosol administrations

were performed as mentioned in Section 2.1. After 0 and 15 min of LPS administration or at the comparable time points in groups without LPS, the cells were harvested. Flow cytometry was then performed as in our previous report (Chen et al., 2012). In addition, harvested RAW264.7 cells were incubated with the primary antibody of CD14 (1:150 dilution, monoclonal CD14 antibody, BD Pharmingen, San Diego, CA, USA) or TLR4/MD2 (1:150 dilution, monoclonal TLR4/MD2 antibody; BD Pharmingen) followed by incubation with an FITC-labelled anti-mouse IgG secondary antibody (1: 200 dilution; BD Pharmingen) to facilitate assaying mCD14 or TLR4/MD2. Subsequently, flow cytometry was performed on a flow cytometer (Cytomic FC50 Flow Cytometer; Beckman Coulter, Inc., Fullerton, CA, USA) to assay LPS-macrophage binding and surface expression of mCD14 and TLR4/MD2.

2.5. Immunofluorescence assay

CD14 plays an essential role in regulating LPS-induced inflammatory response, as CD14 enables the LPS receptor to switch into the “full activation” mode (Jiang et al., 2005). We thus chose to employ immunofluorescence assay to further investigate mCD14 expression. The protocol was adapted from a previous report (Eslani, Movahedan, Afsharkhamseh, Sroussi, & Djalilian, 2014). In brief, RAW 264.7 cells were grown on glass coverslips in 6-well plates (3 × 10⁵ cells per well; Corning) and incubated at 37 °C with 5% CO₂ for 24 h. Then, RAW264.7 cells were treated with PBS, 1.2 mM tyrosol, 100 ng/mL LPS, or 100 ng/mL LPS and 1.2 mM tyrosol [denoted as the PBS, T(1.2), LPS, and LPS + T(1.2) groups, respectively]. LPS and tyrosol were administered as mentioned in Section 2.1. After 15 min of LPS administration or at the comparable time point in the groups without LPS, the cells were harvested. The cells were then fixed with 2% paraformaldehyde (Sigma-Aldrich) for 15 min, followed by washing for 5 min with 1 × PBS (Sigma-Aldrich) three times. To control nonspecific background binding, the cells were blocked with a solution containing 10 mM PBS and 1% bovine serum albumin (primary antibody diluent; Genemed Biotechnologies, Inc., San Francisco, CA, USA) for 60 min. Subsequently, the cells were incubated with primary antibody for CD14 (1:150 dilution, monoclonal CD14 antibody, BD Pharmingen) for 60 min, followed by incubation with fluorescence-conjugated secondary antibody (donkey anti-goat IgG secondary antibody conjugated with Alexa Fluor 594 dye; Thermo Fischer Scientific, Waltham, MA, USA). After washing, nuclei were counterstained with diamidino-2-phenylindole (0.5 µg/mL, Pierce) for 3 min. The coverslips were then mounted using mounting medium (Sigma-Aldrich) and fluorescence was imaged with a fluorescence microscope (Carl Zeiss, Oberkochen, Germany). In addition, quantitative analysis of the cellular fluorescence intensity was determined by drawing circles around representative 10 cells from at least 5 different fields in each group. Then the fluorescence intensity per cell was reported according to the mean fluorescence value using ImageJ software (freely available at <http://rsb.info.nih.gov/ij/>).

2.6. Immunoprecipitation and immunoblotting assay

Immunoprecipitation of TLR4 followed by immunoblotting assay of MyD88 and TRIF were employed for assaying recruitment of MyD88 and TRIF, as previously reported (Chen et al., 2012; Tsai et al., 2006). RAW264.7 cells were seeded in 10-cm culture dishes (1 × 10⁷ cells per dish; Corning) and incubated at 37 °C with 5% CO₂ for 24 h. Then, RAW264.7 cells were treated with PBS, 1.2 mM tyrosol, 100 ng/mL LPS, or 100 ng/mL LPS and 1.2 mM tyrosol [denoted as the PBS, T(1.2), LPS, and LPS + T(1.2) groups, respectively]. For MyD88 assay, cell cultures were harvested after 15 min of LPS administration or at the comparable time point in groups without LPS. For TRIF assay, cell cultures were harvested after 120 min of LPS administration or at comparable time points in groups without LPS. After harvesting, whole cell lysates were prepared as mentioned in Section 2.3.

For immunoprecipitation, cell lysates (300 µg of protein) were first

incubated with TLR4 antibody (1:100, Invivogen Corp., San Diego, CA, USA) followed by reaction with 50% protein A-agarose beads (Sigma-Aldrich). Then, the immunoprecipitates were collected and immunoblotting assay was performed, as mentioned in Section 2.3. For MyD88 assay, the membranes were incubated with the primary antibody of MyD88 (1:200 dilution; monoclonal anti-MyD88 antibody, Abcam, Cambridge, UK) or Actin (1:5000 dilution; Millipore). For TRIF assay, the membranes were incubated with the primary antibody of TRIF (1:200 dilution; monoclonal anti-TRIF antibody, Abcam) or Actin (1:5000 dilution; Millipore). Second antibody conjugation, protein bands detection and densitometry analyses were then performed, as mentioned in Section 2.3.

2.7. Measuring sCD expression using ELISA

To determine the effects of tyrosol on sCD14 expression, RAW 264.7 cells were seeded in 6-well plates (1×10^6 cells per well; Corning) and incubated at 37°C with 5% CO_2 for 24 h. Then, RAW264.7 cells were treated with PBS, 1.2 mM tyrosol, 100 ng/mL LPS, or 100 ng/mL LPS and 1.2 mM tyrosol [denoted as the PBS, T(1.2), LPS, and LPS + T(1.2) groups, respectively]. Of note, CD14 shedding (i.e., sCD14 production) mediated by proteases is the process of proteolytic cleavage of mCD14 (Pham, 2008; Tada et al., 2002). To elucidate the possible role of proteases, another 3 groups of RAW264.7 cells were treated with 1:100-diluted mixture of protease inhibitors (PIs; Protease Inhibitor Cocktail P8340, Sigma-Aldrich); 100 ng/mL LPS and 1:100-diluted PIs; or 100 ng/mL LPS, 1.2 mM tyrosol, and 1:100-diluted PIs [denoted as the PI, LPS + PI, and LPS + T(1.2) + PI groups, respectively]. LPS and tyrosol administration was performed as mentioned in Section 2.1. The mixture of PIs was administered 5 min before tyrosol administration. At 30 min after LPS administration or at the comparable time point in groups without LPS, the cells were harvested. The freshly harvested culture medium was centrifuged at 500g for 5 min, and the supernatant was collected. The sCD14 concentration of the supernatant from each group was assayed using ELISA. A commercial sCD14 ELISA kit (Enzo Life Sciences, Inc., Farmingdale, NY, USA) was employed and analyses were performed as per the manufacturer's protocol.

2.8. Statistical analysis

One-way analysis of variance and post hoc pairwise comparisons with Tukey's test were applied to determine the differences between the groups. A p of < 0.05 was considered significant. SPSS (version 16.0 for Windows; SPSS Inc., Chicago, IL, USA) was used for statistical analyses.

3. Results

3.1. Tyrosol inhibited LPS-induced IL-6 secretion

Fig. 1A illustrated the data of cell viability ($n = 5$ in each group). The mean CCK-8 absorption of each group were compared with the mean CCK-8 absorption in the PBS group and expressed as the relative absorption. Our data revealed comparable cell viabilities in the PBS, T(1.2), LPS, LPS + T(0.3), LPS + T(0.6), and LPS + T(1.2) groups (all $p > 0.05$).

Fig. 1B illustrated the data of IL-6 ($n = 5$ in each group). Our data revealed that the PBS and T(1.2) groups had low IL-6 concentrations. The IL-6 concentration was significantly higher in the LPS group than in the PBS group ($p < 0.001$). IL-6 concentrations did not differ significantly between the LPS + T(0.3) and LPS group, whereas IL-6 concentrations were significantly lower in the LPS + T(0.6) and LPS + T(1.2) groups than in the LPS group in a dose-dependent manner (all $p < 0.05$).

3.2. Tyrosol inhibited LPS-induced NF- κ B and MAPKs expression

Fig. 2 illustrated the representative gel photography and densitometry analyses data of NF- κ B expression ($n = 4$ in each group). Our data revealed that the nuclear concentrations of p-NF- κ B (Fig. 2A) and the cytosolic concentrations of p-I κ B (Fig. 2B) in the PBS and T(1.2) groups were low. In contrast, the nuclear concentration of p-NF- κ B and the cytosolic concentration of p-I κ B in the LPS group were significantly higher than in the PBS group (both $p < 0.001$). Of note, the nuclear concentration of p-NF- κ B and the cytosolic concentration of p-I κ B in the LPS + T(1.2) group were significantly lower than in the LPS group ($p < 0.001$ and $p = 0.003$, respectively).

Fig. 3 illustrated the representative gel photography (Fig. 3A) and densitometry analyses data (Fig. 3B–D) of MAPKs expression ($n = 4$ in each group). Our data revealed that the concentrations of p-ERK (Fig. 3B), p-JNK (Fig. 3C) and p-p38 MAPK (Fig. 3D) in the PBS and T(1.2) groups were low. As expected, the concentrations of p-ERK, p-JNK and p-p38 MAPK in the LPS group were significantly higher than in the PBS group (all $p < 0.001$). In contrast, the concentrations of p-ERK, p-JNK and p-p38 MAPK in the LPS + T(1.2) group were significantly lower than in the LPS group ($p = 0.027$, 0.020 and 0.007 , respectively).

3.3. Tyrosol inhibited LPS binding to RAW264.7 cells

LPS binding to RAW264.7 cells was measured at 0 and 15 min after LPS administration in the LPS and LPS + T(1.2) groups or at the comparable time points in the PBS and T(1.2) groups through flow cytometry ($n = 4$ in each group). Fig. 4A illustrated the representative density plot and flow cytometry histogram of the PBS group. Fig. 4B illustrated the representative flow cytometry histograms of FITC-LPS measured at 15 min after FITC-LPS administration. The mean fluorescence intensities (MFI) of each group were compared with the 0-minute MFI in the PBS group and expressed as the MFI ratio. Fig. 4C illustrated the data of MFI ratio. The 0- and 15-minute MFI ratios of FITC-LPS in the PBS and T(1.2) groups were comparable (data not shown); similarly, the 0-minute MFI ratios of FITC-LPS in the LPS and PBS groups were comparable. By contrast, the 15-minute MFI ratio of FITC-LPS was significantly higher in the LPS group than in the PBS group ($p < 0.001$). The 0-minute MFI ratios of FITC-LPS in the LPS + T(1.2) and LPS groups were comparable, but the 15-minute MFI ratio of FITC-LPS was significantly lower in the LPS + T(1.2) group than in the LPS group ($p = 0.017$).

3.4. Tyrosol inhibited LPS-induced increases in surface expression of mCD14

Surface expression of mCD14 was measured at 0 and 15 min after LPS administration in the LPS and LPS + T(1.2) groups or at the comparable time points in the PBS and T(1.2) groups through flow cytometry ($n = 4$ in each group). Fig. 5A illustrated the representative density plot and flow cytometry histogram of the PBS group. Fig. 5B illustrated the representative flow cytometry histograms of mCD14 measured at 15 min after LPS administration. The MFI of each group, as mentioned, were compared with the 0-minute MFI in the PBS group and expressed as the MFI ratio. Fig. 5C illustrated the data of MFI ratio. The 0- and 15-minute MFI ratios of mCD14 in the PBS and T(1.2) groups were comparable (data not shown). The 0-minute MFI ratios of mCD14 in the LPS and PBS groups were also comparable. By contrast, the 15-minute MFI ratio of mCD14 was significantly higher in the LPS groups than in the PBS group ($p = 0.002$). The 0-minute MFI ratios of mCD14 in the LPS + T(1.2) and LPS groups were also comparable, but the 15-minute MFI ratio of mCD14 was significantly in the LPS + T(1.2) group lower than in the LPS group ($p = 0.031$). Of note, mCD14 data from immunofluorescence staining assay (Fig. 5D) and the cellular fluorescence intensity assay (Fig. 5E) paralleled those from flow cytometry.

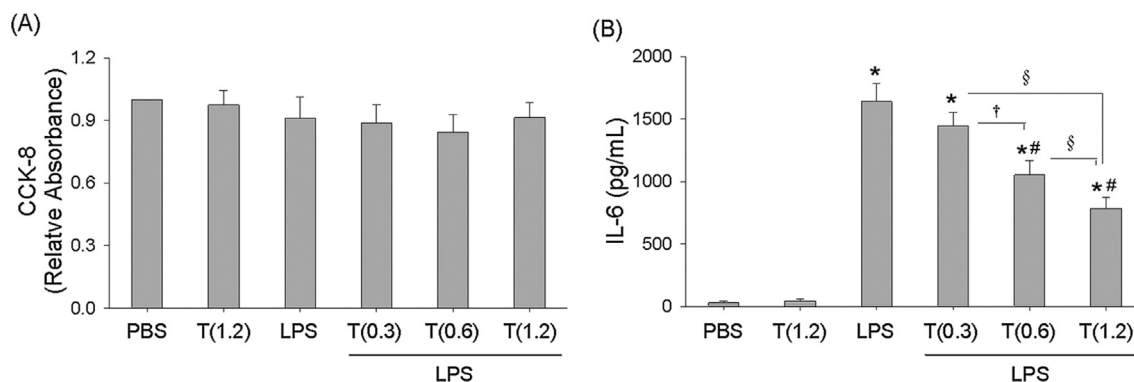


Fig. 1. Cell viability and interleukin-6 (IL-6) concentrations. (A) Cell viability was measured through fluorometric detection of living cell numbers using Cell Counting Kit-8 (CCK-8). (B) IL-6 concentrations in freshly harvested culture media were measured using enzyme-linked immunosorbent assay. RAW264.7 cells were treated with phosphate-buffered saline (PBS), 1.2 mM tyrosol, 100 ng/mL lipopolysaccharide (LPS), 100 ng/mL LPS and 0.3 mM tyrosol, 100 ng/mL LPS and 0.6 mM tyrosol, or 100 ng/mL LPS and 1.2 mM tyrosol [denoted as the PBS, T(1.2), LPS, LPS + T(0.3), LPS + T(0.6), and LPS + T(1.2) groups, respectively]. In the LPS + T(0.3), LPS + T(0.6), and LPS + T(1.2) groups, tyrosol was administered at 60 min before LPS. Cells were harvested at 18 h after LPS administration or at comparable time point in groups without LPS. The mean CCK-8 absorption of each group were compared with the mean CCK-8 absorption in the PBS group and expressed as the relative absorption. Data are mean $\pm$ standard deviations ($n = 5$ in each group). * $p < 0.05$ versus the PBS group. # $p < 0.05$ versus the LPS group. † $p < 0.05$ the LPS + T(0.3) group versus the LPS + T(0.6) group. § $p < 0.05$ the LPS + T(0.3) or LPS + T(0.6) group versus the LPS + T(1.2) group.

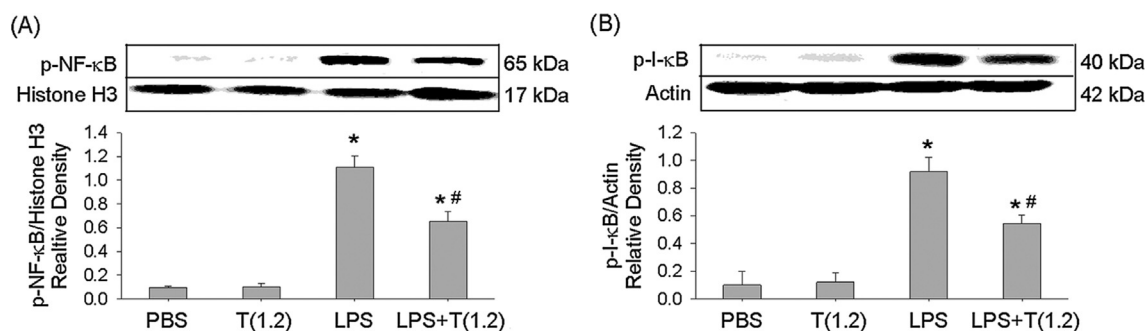


Fig. 2. Effects of tyrosol on modulating nuclear factor- κ B (NF- κ B) expression. NF- κ B expression was measured through immunoblotting assay. (A) Representative gel photograph and the densitometry analysis of the phosphorylated NF- κ B (p-NF- κ B) and Histone H3 (the internal standard) in nuclear extracts measured by immunoblotting assay. (B) Representative gel photograph and the densitometry analysis of the phosphorylated inhibitor- κ B (p-I- κ B) and Actin (the internal standard) in cytosolic extracts measured by immunoblotting assay. RAW264.7 cells were treated with phosphate-buffered saline (PBS), 1.2 mM tyrosol, 100 ng/mL lipopolysaccharide (LPS), or 100 ng/mL LPS and 1.2 mM tyrosol [denoted as the PBS, T(1.2), LPS, and LPS + T(1.2) groups, respectively]. In the LPS + T(1.2) group, tyrosol was administered at 60 min before LPS. Cells were harvested at 30 min after LPS administration or at comparable time point in groups without LPS. Data are mean $\pm$ standard deviations ($n = 4$ in each group). * $p < 0.05$ versus the PBS group. # $p < 0.05$ versus the LPS group.

3.5. Tyrosol mitigated LPS-induced decreases in surface expression of TLR4/MD2

Surface expression of TLR4/MD2 was also measured at 0 and 15 min after LPS administration in the LPS and LPS + T(1.2) groups or at the comparable time points in the PBS and T(1.2) groups through flow cytometry ($n = 4$ in each group). Fig. 6A illustrated the representative density plot and flow cytometry histogram of the PBS group. Fig. 6B illustrated the representative flow cytometry histograms of TLR4/MD2 measured at 15 min after LPS administration. The MFI of each group, also as mentioned, were compared with the 0-minute MFI in the PBS group and expressed as the MFI ratio. Fig. 6C illustrated the MFI ratio data. The 0- and 15-minute MFI ratios of TLR4/MD2 in the PBS and T(1.2) groups were comparable (data not shown). The 0-minute MFI ratios of TLR4/MD2 in the LPS and PBS groups were also comparable. However, the 15-minute MFI ratio of TLR4/MD2 was significantly lower in the LPS group than in the PBS group ($p = 0.009$). The 0-minute MFI ratios of TLR4/MD2 in the LPS + T(1.2) and LPS groups were comparable, whereas the 15-minute MFI ratio of TLR4/MD2 was significantly higher in the LPS + T(1.2) group than in the LPS group ($p = 0.038$).

3.6. Tyrosol inhibited MyD88 and TRIF recruitments

Fig. 7 illustrated the representative gel photograph and densitometry analyses data of MyD88 and TRIF recruitment (Fig. 7A & B, respectively). For MyD88, cell cultures were harvested at 15 min after LPS administration in the LPS and LPS + T(1.2) groups or at the comparable time point in the PBS and T(1.2) groups ($n = 4$ in each group). For TRIF, cell cultures were harvested at 120 min after LPS administration in the LPS and LPS + T(1.2) groups or at the comparable time points in the PBS and T(1.2) groups. After TLR4 immunoprecipitation, immunoblotting assay revealed low MyD88 and TRIF concentrations in the PBS and T(1.2) groups. The MyD88 and TRIF concentrations were significantly higher in the LPS group than in the PBS group (both $p < 0.001$). MyD88 and TRIF concentrations were significantly lower in the LPS + T(1.2) group than in the LPS group ($p = 0.026$ and 0.035 , respectively).

3.7. Effects of tyrosol on enhancing CD14 shedding were counteracted by protease inhibition

For examining CD14 shedding (i.e., proteolytic cleavage of mCD14), cells were harvested at 30 min after LPS administration in the LPS, LPS + T(1.2), LPS + PI, and LPS + T(1.2) + PI groups or at the comparable

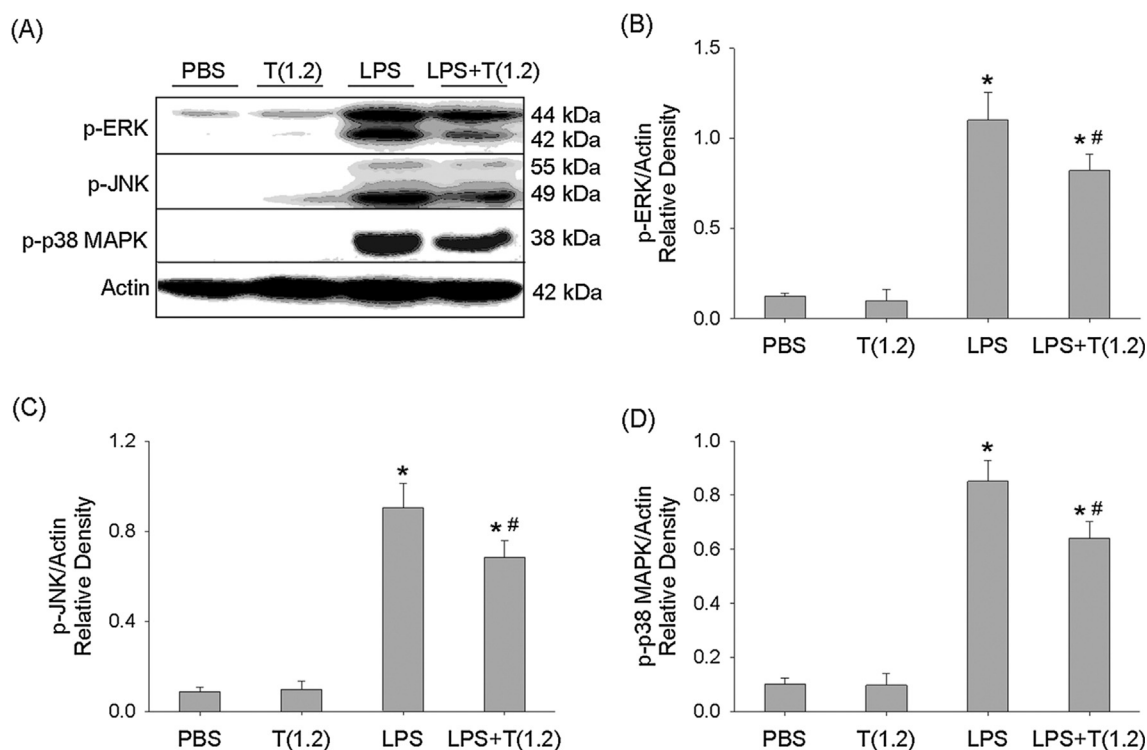


Fig. 3. Effects of tyrosol on modulating mitogen-activated protein kinases (MAPKs) expression. MAPKs expression was measured through immunoblotting assay. (A) Representative gel photograph of immunoblotting assay. (B) Densitometry analysis of the phosphorylated extracellular signal-regulated kinase (p-ERK) and Actin (the internal standard). (C) Densitometry analysis of the phosphorylated c-Jun amino-terminal kinase (p-JNK) and Actin (the internal standard). (D) Densitometry analysis of the phosphorylated p38 MAPK (p-p39 MAPK) and Actin (the internal standard). RAW264.7 cells were treated with phosphate-buffered saline (PBS), 1.2 mM tyrosol, 100 ng/mL lipopolysaccharide (LPS), or 100 ng/mL LPS and 1.2 mM tyrosol [denoted as the PBS, T(1.2), LPS, and LPS + T(1.2) groups, respectively]. In the LPS + T(1.2) group, tyrosol was administered at 60 min before LPS. Cells were harvested at 30 min after LPS administration or at comparable time point in the groups without LPS. Data are mean $\pm$ standard deviations ($n = 4$ in each group). * $p < 0.05$ versus the PBS group. # $p < 0.05$ versus the LPS group.

time point in the PBS, T(1.2), and PI groups. The sCD14 concentrations in the freshly harvested culture media were assayed using ELISA ($n = 4$ in each group). The ELISA data, illustrated in Fig. 8, revealed that the sCD14 concentrations were comparable in the PBS and T(1.2) groups. The sCD14 concentration was significantly higher in the LPS group than

in the PBS group ($p < 0.001$). Similarly, the sCD14 concentration was significantly higher in the LPS + T(1.2) group than in the LPS group ($p = 0.013$). Moreover, the sCD14 concentration was significantly lower in the LPS + T(1.2) + PI group than in the LPS + T(1.2) group ($p < 0.001$). The sCD14 concentration was significantly lower in the PI

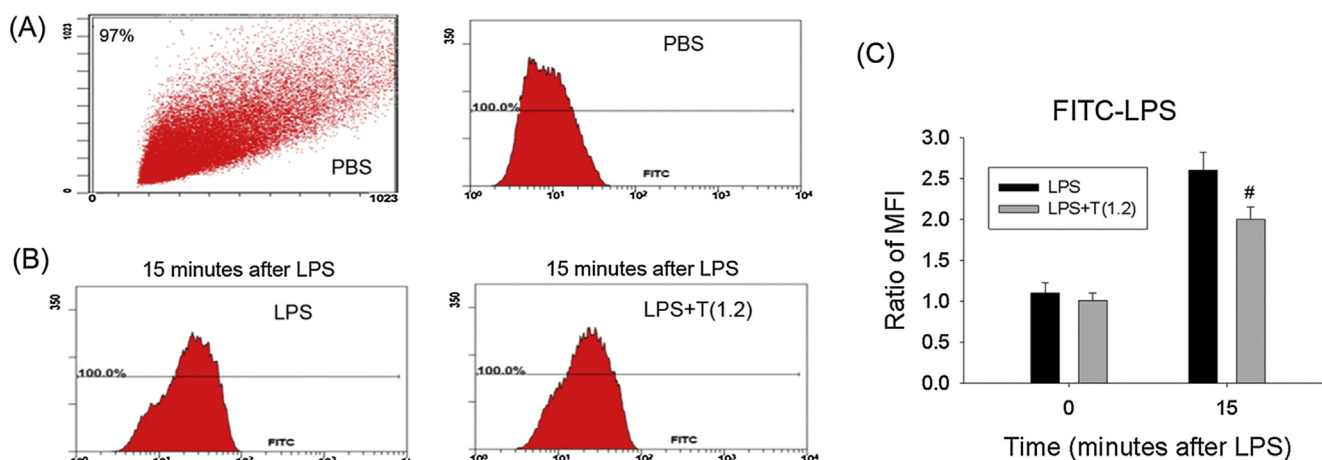


Fig. 4. Effects of tyrosol on modulating lipopolysaccharide (LPS)–macrophages binding. LPS–macrophage binding was assayed through flow cytometry. Fluorescein isothiocyanate-labelled LPS (FITC-LPS) was used to facilitate investigation. (A) Representative density plot and flow cytometry histogram of RAW264.7 cells treated with phosphate-buffered saline (PBS). (B) Representative flow cytometry histogram of FITC-LPS measured at 15 min after FITC-LPS administration. (C) The ratio of mean fluorescence intensities (MFI) of FITC-LPS. RAW264.7 cells were treated with PBS, 1000 ng/mL FITC-LPS, and 1000 ng/mL FITC-LPS and 1.2 mM tyrosol [denoted as the PBS, LPS, and LPS + T(1.2) groups, respectively]. In the LPS + T(1.2) group, tyrosol was administered at 60 min before LPS. Cells were harvested at 0 and 15 min after LPS administration or at comparable time points in groups without LPS. The MFI of each group was compared with the 0-minute MFI of the PBS group and expressed as the ratio of MFI. Data are mean $\pm$ standard deviations ($n = 4$ in each group). # $p < 0.05$ versus the LPS group. One group of RAW264.7 cells was treated with 1.2 mM tyrosol [denoted as the T(1.2) group]. However, we did not report the data of the T(1.2) group in this figure due to space limitations.

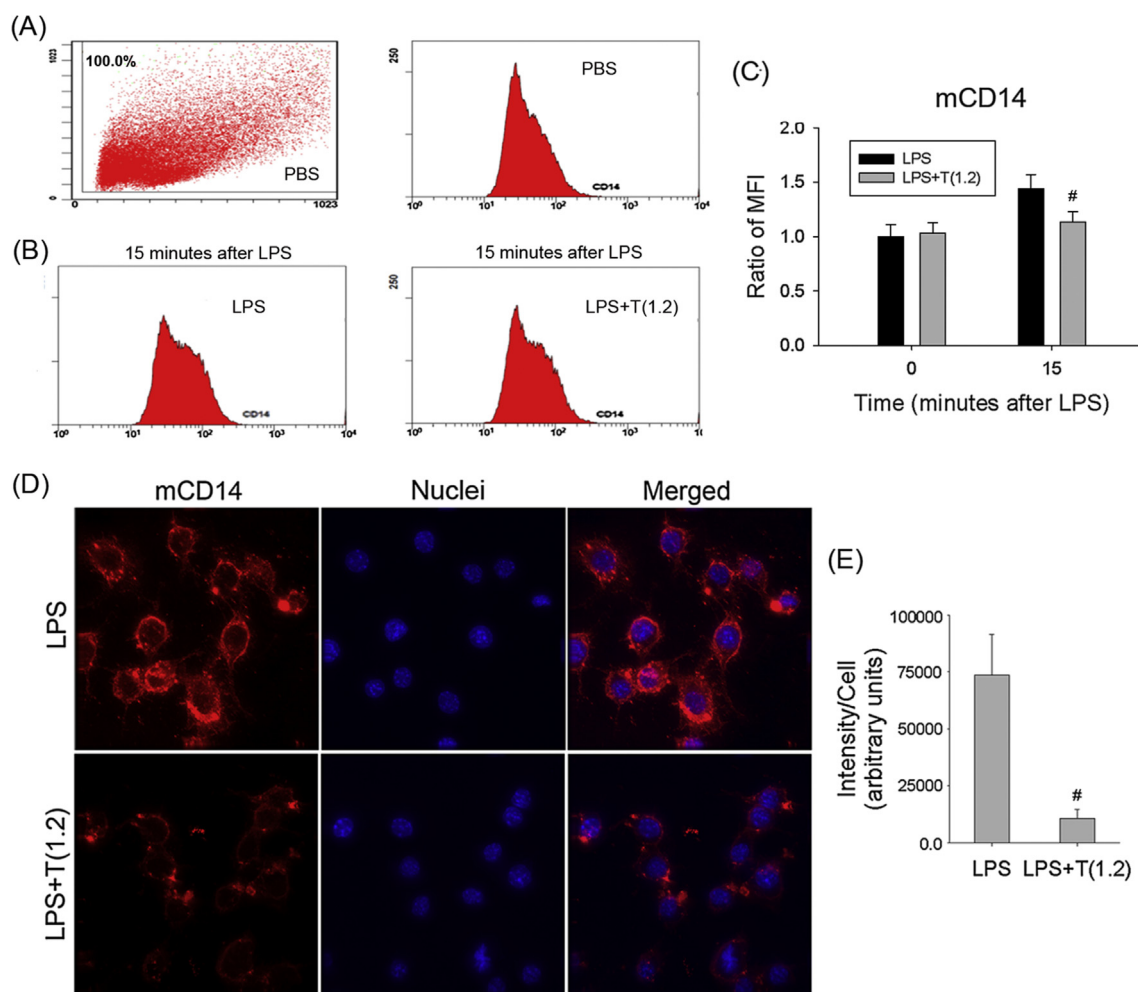


Fig. 5. Effects of tyrosol on modulating membrane-bound cluster of differentiation 14 (mCD14) expression. mCD14 expression was assayed through flow cytometry and immunofluorescent staining assay. (A) Representative density plot and flow cytometry histogram of RAW264.7 cells treated with phosphate-buffered saline (PBS). (B) Representative flow cytometry histograms of mCD14 measured at 15 min after lipopolysaccharide (LPS) administration. (C) The mean fluorescence intensity (MFI) ratios of mCD14. RAW264.7 cells were treated with PBS, 100 ng/mL LPS, or 100 ng/mL LPS and 1.2 mM tyrosol [denoted as the PBS, LPS, and LPS + T(1.2) groups, respectively]. In the LPS + T(1.2) group, tyrosol was administered at 60 min before LPS. Cells were harvested at 0 and 15 min after LPS administration or at comparable time points in groups without LPS. The MFI of each group was compared with the 0-minute MFI of the PBS group and expressed as MFI ratios. (D) Representative images of immunofluorescent staining assay of mCD14 measured at 15 min after LPS administration. mCD14 was stained with Alexa Fluor 594 dye-conjugated antibody. Nuclei were counterstained with diamidino-2-phenylindole. (E) Quantitative analysis of the cellular fluorescence intensity of the images of immunofluorescent staining assay. Data are mean $\pm$ standard deviations ($n = 4$ in each group). [#] $p < 0.05$ versus the LPS group. One group of RAW264.7 cells was treated with 1.2 mM tyrosol [denoted as the T(1.2) group]. However, we did not report the data of the T(1.2) group in this figure due to space limitations.

group than in the PBS group ($p = 0.037$). In addition, the sCD14 concentration was significantly lower in the LPS + PI group than in the LPS group ($p < 0.001$).

4. Discussion

Studies have highlighted the potent anti-inflammatory capacity of tyrosol (Giovannini et al., 2002; Kim et al., 2017; Lu et al., 2013; Sato et al., 2016; Serra et al., 2017; Wang et al., 2017). Findings from this study demonstrated that tyrosol can inhibit inflammatory cytokine IL-6 upregulation in a dose-dependent manner. These data are consistent with previous data that tyrosol can inhibit LPS-induced cytokine production in human peripheral blood mononuclear cells (Giovannini et al., 2002). In addition, the effects of tyrosol on mitigating NF- κ B and MAPKs upregulation were also observed by this study. These data provide clear evidence to support the concept that NF- κ B and MAPKs are involved in mediating the anti-inflammation effects of tyrosol (Sato et al., 2016; Wang et al., 2017). This study further elucidated the effects of tyrosol on modulating the upstream pathway regulating LPS-induced

inflammatory signalling pathways. In activated macrophages, this study confirmed that tyrosol (1) inhibits LPS-macrophage binding, (2) mitigates mCD14 activation by enhancing CD14 shedding, (3) mitigates TLR4/MD2 expression, and (4) reduces MyD88 and TRIF recruitments. This is the first evidence highlighting the anti-inflammatory effects of tyrosol involving its mediation in the upstream regulatory pathways. The current findings contribute to the knowledge regarding tyrosol and can potentially expand the scope of this natural polyphenol's application.

As noted in previous study, LPS-macrophage binding is mediated by CD14 (Jerala, 2007; Schumann, 1992). Moreover, the activation of the TLR4/MD2 complex and subsequent recruitment of MyD88 and TRIF is also mediated by CD14 (Fitzgerald et al., 2004; Jerala, 2007; Pålsson-McDermott and O'Neill, 2004; Park & Lee, 2013; Schumann, 1992). This study highlighted the central role of CD14 in mediating the LPS-induced inflammatory signalling pathways; notably, tyrosol enhances CD14 shedding and subsequently mitigates mCD14 upregulation. Therefore, in LPS-activated macrophages, tyrosol likely acts by enhancing CD14 shedding to inhibit CD14-mediated LPS-macrophage

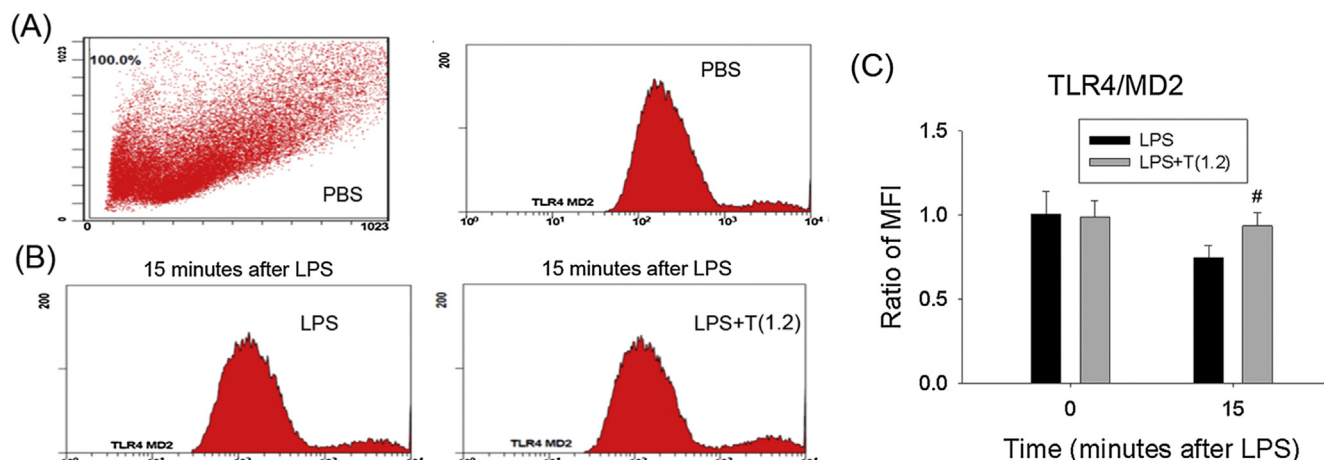


Fig. 6. Effects of tyrosol on modulating surface expression of toll-like receptor 4 (TLR4)/myeloid differentiation factor 2 (MD2). Surface expression of TLR4/MD2 expression was assayed through flow cytometry. (A) Representative density plot and flow cytometry histogram of RAW264.7 cells treated with phosphate-buffered saline (PBS). (B) Representative flow cytometry histograms of TLR4/MD2 measured at 15 min after lipopolysaccharide (LPS) administration. (C) The mean fluorescence intensity (MFI) ratios of TLR4/MD2. RAW264.7 cells were treated with phosphate-buffered saline (PBS), 100 ng/mL LPS, or 100 ng/mL LPS and 1.2 mM tyrosol [denoted as the PBS, LPS, and LPS + T(1.2) groups, respectively]. In the LPS + T(1.2) group, tyrosol was administered at 60 min before LPS. Cells were harvested at 0 and 15 min after administration of LPS or at comparable time points in groups without LPS. The MFI of each group was compared with the 0-minute MFI of the PBS group and expressed as MFI ratios. Data are mean $\pm$ standard deviations ($n = 4$ in each group). [#] $p < 0.05$ versus the LPS group. One group of RAW264.7 cells was treated with 1.2 mM tyrosol [denoted as the T(1.2) group]. However, we did not report the data of the T(1.2) group in this figure due to space limitations.

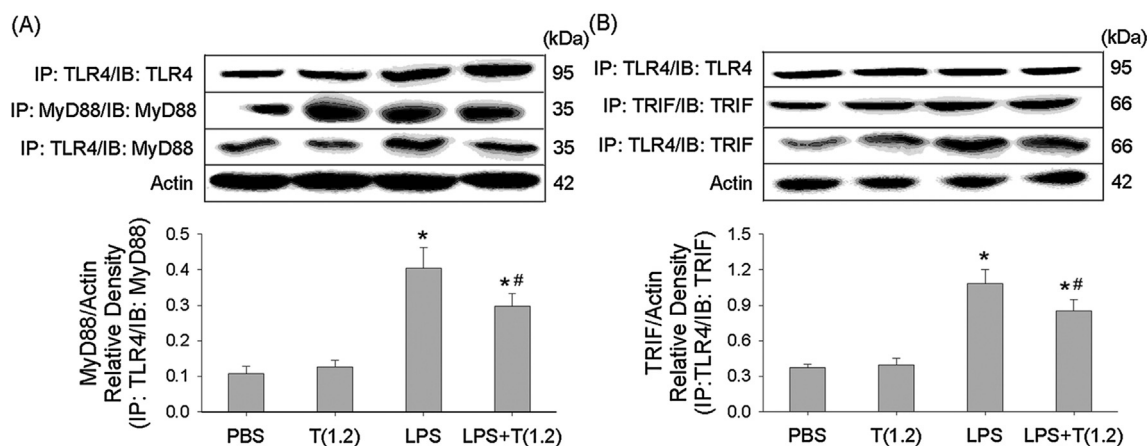


Fig. 7. Effects of tyrosol on modulating recruitment of myeloid differentiation primary response gene 88 (MyD88) and TIR-domain-containing adapter-inducing interferon- β (TRIF). MyD88 and TRIF recruitments were measured through immunoprecipitation (IP) with anti-Toll-like receptor 4 (TLR4) antibody followed by immunoblotting (IB) with antibodies for MyD88 or TRIF. To serve as controls, TLR4, MyD88, and TRIF were individually immunoprecipitated and then immunoblotted. (A) Representative gels and densitometry analyses of MyD88 and Actin (the internal standard). (B) Representative gels and densitometry analyses of TRIF and Actin (the internal standard). RAW264.7 cells were treated with phosphate-buffered saline (PBS), 1.2 mM tyrosol, 100 ng/mL LPS, or 100 ng/mL LPS and 1.2 mM tyrosol [denoted as the PBS, T(1.2), LPS, and LPS + T(1.2) groups, respectively]. In the LPS + T(1.2) group, tyrosol was administered at 60 min before LPS. For MyD88 assay, cell cultures were harvested after 15 min of LPS administration or at comparable time points in groups without LPS. For TRIF assay, cell cultures were harvested after 120 min of LPS administration or at comparable time points in groups without LPS. Data are mean $\pm$ standard deviations ($n = 4$ in each group). ^{*} $p < 0.05$ versus the PBS group. [#] $p < 0.05$ LPS + T(1.2) group versus the LPS group.

binding and subsequently mitigate TLR4/MD2 expression and reduce MyD88 and TRIF recruitments. MyD88 and TRIF recruitments lead to upregulation of NF- κ B and MAPKs expression (Christman et al., 1998; Kristof et al., 2001); thus, it is reasonable that this study as well as those previous studies (Sato et al., 2016; Wang et al., 2017) observed the effects of tyrosol on mitigating upregulation of NF- κ B and MAPK.

Our data that the sCD14 concentrations in the LPS + T(1.2) group were significantly higher than in the LPS group demonstrated that tyrosol, as mentioned, enhances CD14 shedding in LPS-stimulated macrophages. As CD14 shedding is the process of proteolytic cleavage of mCD14 (Pham, 2008; Tada et al., 2002), these findings provide clear evidence to support the concept that tyrosol can enhance proteolytic cleavage of mCD14. As mentioned, proteolytic cleavage of CD14 is

mediated by proteases (Pham, 2008; Tada et al., 2002). Our data that the effects of tyrosol on enhancing CD14 shedding could be counteracted by nonselective protease inhibition further support the concept that tyrosol can act by increasing protease activity to enhance proteolytic cleavage of mCD14 in LPS-activated macrophages. Cysteine and serine proteases are key proteases participating in this proteolytic cleavage (Pham, 2008; Tada et al., 2002). The mixture of protease inhibitors employed in this study (i.e., Protease Inhibitor Cocktail from Sigma-Aldrich) included inhibitors of both cysteine protease and serine protease, data from this study thus suggest the involvement of cysteine protease and/or serine protease in mediating the effects of tyrosol on enhancing CD14 shedding. However, proteases other than cysteine and serine proteases (e.g., metalloproteinase) have also been reported to

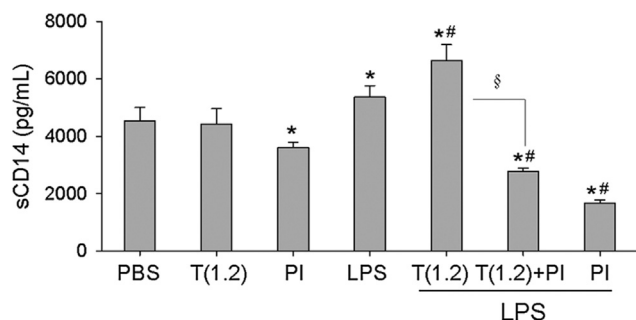


Fig. 8. Effects of tyrosol on modulating soluble cluster of differentiation 14 (sCD14) production and the possible roles of proteases. sCD14 concentrations in freshly harvested culture media were measured using enzyme-linked immunosorbent assay. RAW264.7 cells were treated with phosphate-buffered saline (PBS), 1.2 mM tyrosol, 100 ng/mL LPS, or 100 ng/mL LPS and 1.2 mM tyrosol [denoted as the PBS, T(1.2), LPS, and LPS + T(1.2) groups, respectively]. To elucidate the possible role of proteases, another three groups of RAW264.7 cells were treated with a mixture of protease inhibitors (PIs); 100 ng/mL LPS and PIs; or 100 ng/mL LPS, 1.2 mM tyrosol and PIs [denoted as the PI, LPS + PI, and LPS + T(1.2) + PI groups, respectively]. In the LPS + T(1.2) group, tyrosol was administered at 60 min before LPS. In the LPS + T(1.2) + PI group, PIs were administered at 5 min before tyrosol. Cells were harvested at 30 min after LPS administration or at comparable time points in groups without LPS. Data are mean $\pm$ standard deviations ($n = 4$ in each group). * $p < 0.05$ versus the PBS group. # $p < 0.05$ the LPS + T(1.2), LPS + T(1.2) + PI, or LPS + PI group versus the LPS group. § $p < 0.05$ the LPS + T(1.2) + PI group versus the LPS + T(1.2) group.

participate in proteolytic cleavage of CD14 (Senft, Korfhagen, Whitsett, Shapiro, & Levine, 2005). As the PI mixture used in this study did not include a metalloproteinase inhibitor, the involvement of metalloproteinases in mediating the effects of tyrosol on CD14 shedding enhancement remains unknown. Thus, additional studies using specific cysteine protease, serine protease, or metalloproteinase inhibitors are warranted. In addition, the mechanisms underlying the effects of tyrosol on increasing cysteine and serine protease activities remain unknown. Previous study did indicate that reactive oxygen species are involved in modulating the activity of proteases in the process of inflammation (Bryniarski, Maresz, Szczepanik, Ptak, & Ptak, 2003). As tyrosol possesses potent free radical scavenging capacity (Warleta et al., 2011), we thus speculate that tyrosol might act by decreasing reactive oxygen species to exert its effects on enhancing protease activity. More studies are needed before further conclusion can be drawn.

It is established that binding of LPS to TLR4/MD2 induces TLR4/MD2 activation, including initial TLR4/MD2 dimerization and subsequent TLR4 endocytosis (Fitzgerald et al., 2004; Jerala, 2007; Pålsson-McDermott and O'Neill, 2004; Park & Lee, 2013; Schumann, 1992). As mentioned, the initial dimerization of TLR4/MD2 at the cell surface induces MyD88 recruitment (Kagan & Medzhitov, 2006) and subsequently induces NF- κ B and MAPKs activation and downstream inflammatory molecules upregulation (Christman et al., 1998; Kristof et al., 2001). Although this study did not directly investigate the effects of tyrosol on modulating TLR4/MD2 dimerization, our data did confirm that tyrosol can mitigate MyD88 recruitment, decrease NF- κ B and MAPKs activation and inhibit inflammatory cytokine IL-6 production. Judging from these data, we thus believe that tyrosol can exert significant effects on inhibiting TLR4/MD2 dimerization in LPS-stimulated macrophages. In addition, TLR4/MD2 dimerization leads to TLR4 endocytosis (Fitzgerald et al., 2004; Jerala, 2007; Pålsson-McDermott and O'Neill, 2004; Park & Lee, 2013; Schumann, 1992) and subsequent TRIF recruitment (O'Neill et al., 2003; Yamamoto et al., 2003). Flow cytometry is a sensitive assay to detect surface expression of TLR4/MD2 (Chen et al., 2012; Tan, Zanoni, Cullen, Goodman, & Kagan, 2015). Data from this study revealed that LPS induces a significant decrease in the fluorescence signal intensity of TLR4/MD2, as the 15-minute MFI

ratio of TLR4/MD2 in the LPS group was significantly lower than in the PBS group. These data demonstrated the effects of LPS on decreasing surface expression of TLR4/MD2 in stimulated macrophages. Moreover, our data revealed that the 15-minute MFI ratio of TLR4/MD2 in the LPS + T(1.2) group was significantly higher than in the LPS group. These data further demonstrated the effects of tyrosol on mitigating LPS-induced decreases in surface expression of TLR4/MD2 in stimulated macrophages. As loss of cell surface expression of TLR4/MD2 is used as readout for TLR4 endocytosis measured through flow cytometry (Chen et al., 2012; Tan et al., 2015), our data support the concept that tyrosol can mitigate LPS-induced TLR4 endocytosis in stimulated macrophages. Collectively, these data indicate the effects of tyrosol on decreasing TLR4/MD2 activation, including TLR4/MD2 dimerization and TLR4 endocytosis, in LPS-stimulated macrophages.

TLR4 alone cannot recognise LPS (Maeshima & Fernandez, 2013; Shimazu et al., 1999). By contrast, MD2 consists of one large open-binding hydrophobic pocket, which can accommodate LPS (Shimazu et al., 1999). Therefore, forming a stable 1:1 complex with MD2 (i.e., the TLR4/MD2 complex) is essential for TLR4 to recognise LPS. On contact with LPS transferred by CD14, the hydrophobic pocket of MD2 can accommodate up to five fatty acid chains of LPS (Billod, Lacetera, Guzmán-Caldentey, & Martín-Santamaría, 2016). The sixth fatty acid chain of LPS protrudes from MD2 pocket and forms a TLR4/MD2 dimerization interface (Billod et al., 2016). Based on its effects on transferring LPS to MD2, CD14 is essential in mediating TLR4/MD2 dimerization. In addition, TLR4 is highly dynamic and thus it moves freely in the plasma membrane (Klein et al., 2015). In the presence of LPS, an immobilised fraction of TLR4 in the plasma membrane forms (Klein et al., 2015). This immobilised fraction appears in small punctate structures in the plasma membrane containing CD14/LPS (Klein et al., 2015). Forming the punctuate structures in the plasma membrane is believed to prepare TLR4 for endocytosis (Klein et al., 2015). The presence of CD14/LPS in these structures highlights the crucial roles of CD14 in mediating TLR4 endocytosis. As findings from this study demonstrated the effects of tyrosol on inhibiting both CD14 upregulation and TLR4/MD2 expression, we thus believe that tyrosol may very likely act by inhibiting CD14 to exert its effects on mitigating TLR4/MD2 expression.

The potent anti-inflammatory capacity of tyrosol is confirmed by this study. These in vitro data support the concept that tyrosol has therapeutic potential for inflammatory disease. The main source of tyrosol is the Mediterranean diet (Miró-Casas et al., 2001). Epidemiological studies showed polyphenols in the Mediterranean diet reduce the risk of coronary heart disease mortality and morbidity (Miró-Casas et al., 2001). Bioavailability of tyrosol is the main issue for applying the findings from bench to bedside. Tyrosol is absorbed in a dose-dependent manner with intake of moderate doses of phenol-rich natural sources (e.g., olive oil) (Covas et al., 2003; Miró-Casas et al., 2001). Tyrosol is relatively stable in gastro-intestinal transit and mainly absorbed in small intestine (Visser, Zock, Roodenburg, Leenen, & Katan, 2002). Only a fraction (approximate 20–22%) of tyrosol is excreted in urine within 24 h after ingestion (Covas et al., 2003; Miró-Casas et al., 2001; Visser et al., 2002; Visioli et al., 2000). Collectively, these data indicate that tyrosol from phenol-rich natural sources can be absorbed and used effectively by the human body (Covas et al., 2003; Miró-Casas et al., 2001; Visser et al., 2002; Visioli et al., 2000). In vivo data derived from rats further demonstrated a very high oral bioavailability of tyrosol (up to 98%) (Tuck, Freeman, Hayball, Stretch, & Stupans, 2001). Natural sources of tyrosol, as mentioned, include olive oil, red wine, and white wine (Covas et al., 2003; Miró-Casas et al., 2001). Previous data indicate that oral bioavailability of compounds can be different when administered as aqueous and oil solutions (Lyons, Charman, Miller, & Porter, 2000). This concept is confirmed by previous data that the estimated oral bioavailability of tyrosol can drop from 98% to 75% if tyrosol is orally administered in an aqueous solution rather than an olive oil solution (Tuck et al., 2001). The other issue

needs to be resolved before these findings can be applied from bench to bedside. The highest tyrosol dose investigated here was 1.2 mM (= 198.95 µg/mL). The daily intake of virgin olive oil, a major natural source of tyrosol, in a typical Mediterranean diet is estimated to be approximately 25 mL, which contains approximately 860 µg of tyrosol (Miró-Casas et al., 2001). With high bioavailability and approximate 20% of ingested tyrosol is excreted in urine within 24 h, as mentioned; thus, with a daily intake of 25 mL of virgin olive oil, approximately 700 µg of tyrosol is absorbed. Therefore, the daily tyrosol intake through ingestion of natural sources in a standard Mediterranean diet is much greater than the dose investigated in this study. Thus, the doses of tyrosol investigated in this study are physiologically relevant and achievable in vivo. Our findings could thus have profound clinical implications.

This study also confirmed the potent effects of tyrosol on inhibiting endotoxin-induced inflammatory response. The data indicated the therapeutic potential of tyrosol against clinical conditions related to endotoxin-induced inflammation (e.g., sepsis). This concept is supported by in vivo data revealing that tyrosol can mitigate endotoxin-induced organ injuries and increase survival in rodents with endotoxemia (Kim et al., 2017; Lu et al., 2013). Thus, tyrosol may have therapeutic effects and serve as a therapeutic adjuvant against sepsis; it may also mitigate organ injuries and improve outcomes in patients with endotoxin-induced sepsis. Additional clinical studies are required before our findings translate into clinical use. In addition, our results seem to confirm that regular intake of olive oil (as in a typical Mediterranean diet) may provide beneficial effects and boost anti-inflammatory capacity against conditions related to endotoxin-induced inflammatory response. These data further support regular olive oil intake as part of a healthy diet. Moreover, natural sources of tyrosol also include red and white wines (Covas et al., 2003; Miró-Casas et al., 2001). As mentioned, previous data indicated lower oral bioavailability of tyrosol with administration in an aqueous solution (Tuck et al., 2001). It is thus likely that red and white wines are less effective natural sources of tyrosol comparing to olive oils. However, red and white wines can still be good natural sources of tyrosol, as substantial amounts of tyrosol can be absorbed with an estimated oral bioavailability of approximate 75% (Tuck et al., 2001). Thus, we further speculate that regular intake of these natural sources of tyrosol may provide beneficial effects and mitigate endotoxemia-induced organ injuries. However, additional relevant clinical studies are required before the beneficial effects of any natural tyrosol sources can be concluded.

CD14 upregulation and subsequent inflammation are the crucial mechanisms mediating the development of sepsis induced by endotoxin (Fitzgerald et al., 2004; Jerala, 2007; Pålsson-McDermott and O'Neill, 2004; Park & Lee, 2013; Schumann, 1992). Similar to endotoxin-induced sepsis, CD14 and subsequent inflammation has been linked to several other diseases not mediated by the presence of endotoxin. For instance, CD14 upregulation and inflammation have been noted in cardiovascular diseases, including hypertension, atherosclerosis, coronary heart disease, and chronic heart failure (Anker et al., 1997; Arroyo-Espliguero, Avanzas, Jeffery, & Kaski, 2004). Insulin resistance and obesity may also be associated with CD14 upregulation and inflammation (De Loera-Rodriguez et al., 2014; Fernández-Real et al., 2011). The effects of tyrosol on inhibiting CD14 upregulation and mitigating inflammation were confirmed in this study. Thus, tyrosol likely has beneficial effects against hypertension, atherosclerosis, coronary heart disease, chronic heart failure, insulin resistance, and obesity by modulating CD14 upregulation and inhibiting inflammation. Regular intake of natural sources of tyrosol may thus have beneficial effects against these diseases. In vivo studies, employing animal models and humans, are thus warranted.

Similar to tyrosol, hydroxytyrosol, a major polyphenol presents in olive oil (Covas et al., 2003; Miró-Casas et al., 2001), may also possess potent anti-inflammatory effects (Zhang, Cao, & Zhong, 2009; Yang

et al., 2017). Hydroxytyrosol could inhibit cytokine production in LPS-stimulated human monocytic cells (Zhang et al., 2009) and mitigate acute lung injury in rodents with endotoxemia (Yang et al., 2017). In addition to tyrosol and hydroxytyrosol, the anti-inflammatory effects of many other polyphenols have been reported. For instance, the citrus flavonoids hesperidin and hesperetin can mitigate inflammation through inhibiting NF-κB, inducible nitric oxide synthase, and cyclooxygenase 2 (Parhiz, Roohbakhsh, Soltani, Rezaee, & Iranshahi, 2015). Common polyphenols in fruits, vegetables, and grains (e.g., flavonol quercetin and anthocyanins) can exert substantial effects on modulating endotoxin-induced inflammation (Wang & Mazza, 2002). In addition, polyphenols from cocoa and nuts can mitigate inflammation and exert cardiovascular protective effects (Natsume, 2018; Rusu, Gheldiu, Mocan, Vlase, & Popa, 2018). Because polyphenols have potent anti-inflammatory properties, regular intake of phenol-rich natural sources (e.g., olive oil, red wine, white wine, citrus, fruits, vegetables, grains, cocoa, nuts) may boost the endogenous anti-inflammatory capacity of the human body. Polyphenols thus have great therapeutic potential. Novel therapies based on polyphenols that may provide natural alternatives for preventing or treating inflammation-related diseases should be investigated.

The above-mentioned effects of tyrosol and the possible mechanisms are summarized in Fig. 9. Nevertheless, this study has several limitations. First, we used a widely used immortalised macrophage activation model, namely the RAW264.7 cell line treated with LPS; however, the corresponding effects on primary cell cultures remain unknown. Second, this was a cellular study; whether tyrosol can exert similar effects in intact animals remains unknown. Third, tyrosol was administered 60 min before LPS; therefore, only the 'pretreatment' effects of tyrosol were observed in this study. Whether tyrosol exerts similar effects after administration at different time points (e.g., cotreatment or posttreatment) remains unclear. Fourth, tyrosol mitigates mCD14 activation by enhancing protease-mediated cleavage of CD14. However, whether tyrosol directly activates the proteases remains unclear. Fifth, LPS exposure for 15 min could induce significant mCD14 expression on RAW264.7 cells; this is consistent with previous findings that LPS can significantly induce monocyte mCD14 upregulation (~50% increase) within 30 min (Marchant, Duchow, Delville, & Goldman, 1992). Because the reaction time was short, the underlying mechanisms may involve translocation of an intracellular pool of CD14 rather than protein synthesis (Marchant et al., 1992). In addition, the current findings indicated that tyrosol pretreatment can enhance proteolytic cleavage of mCD14. The effects of LPS on increasing mCD14 expression are likely negated by enhanced proteolytic cleavage of mCD14 induced by tyrosol pretreatment. Thus, our data indicate that the mCD14 expression was significantly lower in the LPS + T(1.2) group than that in the LPS group. Nevertheless, this study did not investigate mCD14 source, and therefore, whether the increased mCD14 expression in LPS-stimulated RAW264.7 cells is derived from translocation of an intracellular pool of CD14 remains unclear. Sixth, the LPS-induced signalling pathway is mediated by both the CD14-dependent and -independent pathways (Płóciennikowska, Hromada-Judycka, Borzęcka, & Kwiatkowska, 2015). Our findings revealed that the CD14-dependent pathway is involved in mediating the effects of tyrosol on inhibiting the LPS-induced signalling pathway. However, whether this effect also involves the CD14-independent pathway requires further study. Finally, although this study indicates that tyrosol can inhibit TLR4/MD2 expression in LPS-stimulated macrophages, direct evidence delineating the effects of tyrosol on inhibiting TLR4/MD2 dimerization and endocytosis remains lacking.

5. Conclusion

Tyrosol can inhibit LPS-macrophage binding, mitigate mCD14 and TLR4/MD2 expression, and reduce MyD88 and TRIF recruitments in LPS-activated macrophages. The mechanisms may involve tyrosol-

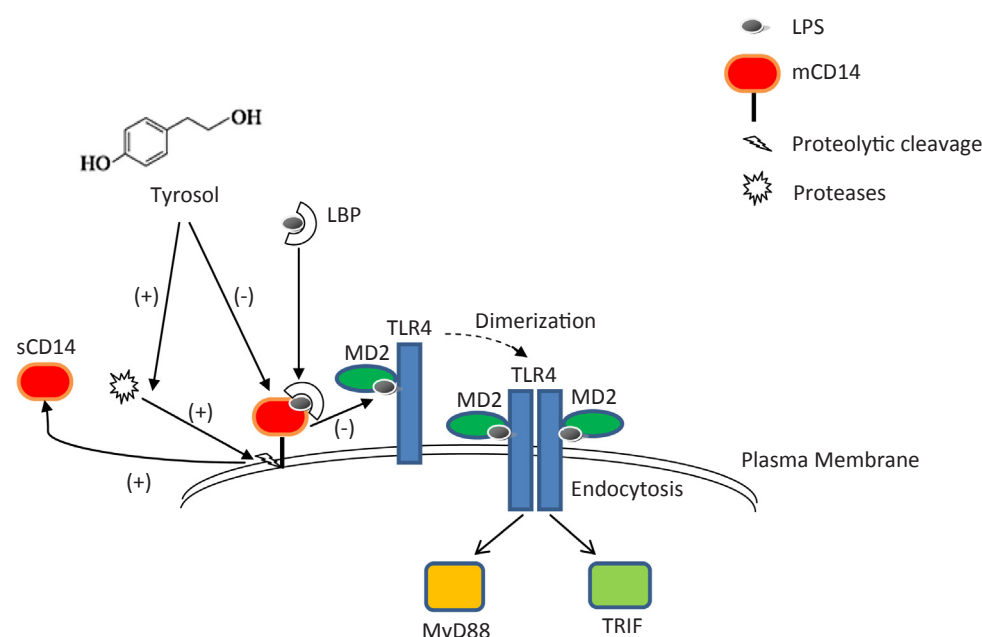


Fig. 9. Schematic illustration of the effects of tyrosol on inhibiting cluster of differentiation 14 (CD14)-mediated lipopolysaccharide (LPS)-macrophage binding and mitigating expression of toll-like receptor 4 (TLR4)/myeloid differentiation factor 2 (MD2) complex and recruitment of myeloid differentiation primary response gene 88 (MyD88) and TIR-domain-containing adapter-inducing interferon-beta (TRIF) in LPS-activated macrophages. The mechanisms may involve the effects of tyrosol on enhancing proteolytic cleavage of membrane-bound CD14 (mCD14), i.e., CD14 shedding, mediated by proteases. sCD14: soluble CD14, the product of CD14 shedding. LBP: LPS binding protein. (+): enhance or increase. (-): inhibit or decrease.

enhanced proteolytic cleavage of mCD14.

Conflict of interests

The authors declared that there is no conflict of interest.

Ethical statements

This research did not include any human subjects and animal experiments.

Disclosure

The authors report no proprietary or commercial interest in any product mentioned or concept discussed in this article.

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